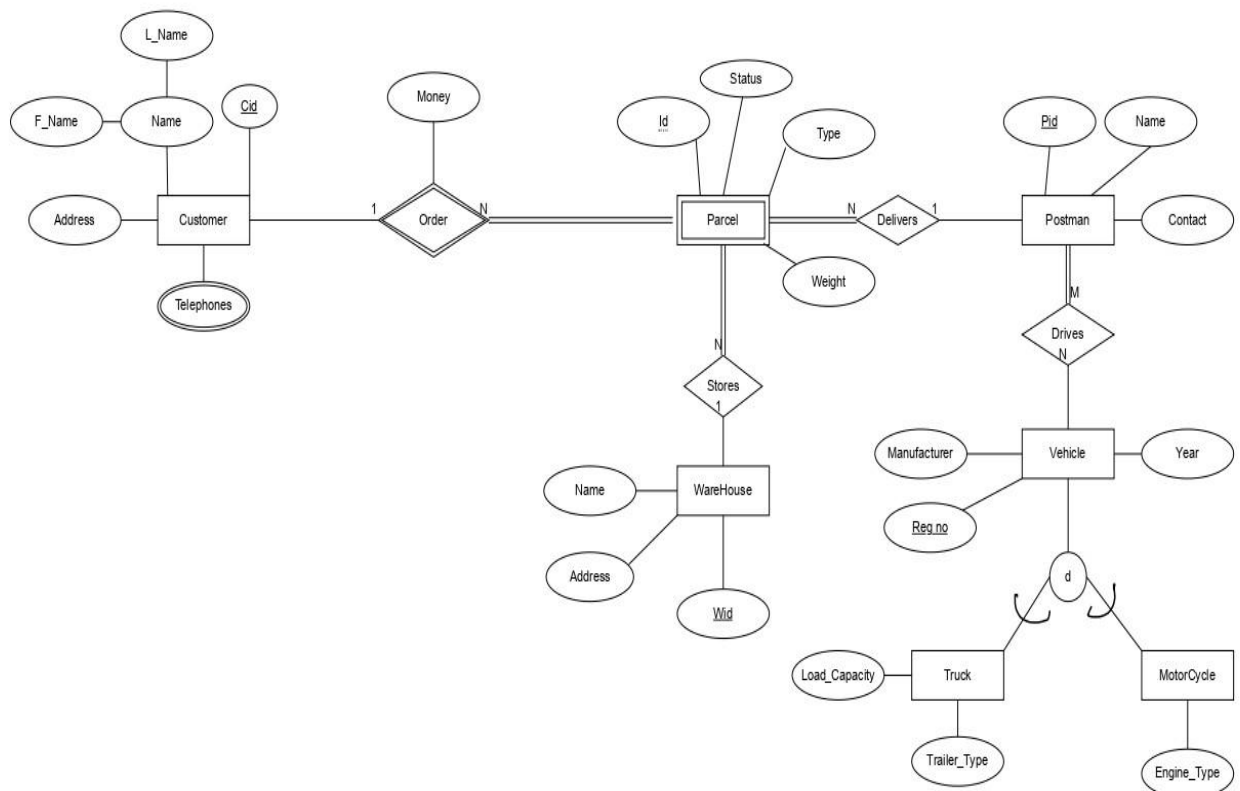


# Report

## UoD Recap:

The Parcel Delivery System records essential details of customers, parcels, warehouses, postmen, and vehicles involved in courier operations. Customers can place multiple parcel orders, and each parcel is tracked by its weight, status, and type. Parcels are stored in warehouses and delivered by postmen who use either trucks or motorcycles. Vehicles are defined by their type-specific attributes. The system allows staff to update parcel status (e.g., in transit, delivered), remove delivered parcels from records, check how many parcels are stored in each warehouse, and track which postman delivered which parcel.

## ERD:



## BCNF-normalized schema:

### 1.Customer:

- Customer [Cid :INT , F\_Name: VARCHAR(50), L\_Name: VARCHAR(50),Address: VARCHAR(100)]
- Primary key: Cid
- Foreign keys: None

### 2.Parcel:

- Parcel [Id : INT ,Cid:INT , Status: VARCHAR(50), Type: VARCHAR(50), Weight: DECIMAL(5,2), Money: DECIMAL(6,2), Wid: INT, Pid: INT]
- Primary key: (Id,Cid)
- Foreign keys:
  - Cid → Customer.Cid
  - Wid → WareHouse.Wid
  - Pid → Postman.Pid

### 3.WareHouse:

- WareHouse [Wid : INT , Address: VARCHAR(200), Name: VARCHAR(100)]
- Primary key: Wid
- Foreign keys: None

### 4.Postman:

- Postman [Pid : INT , Name: VARCHAR(100),Contact: VARCHAR(20)]
- Primary key: Pid
- Foreign keys:None

### 5.Vehicle:

- Vehicle [Reg\_no : VARCHAR(50) , Manufacturer: VARCHAR(50),Year: INT]
- Primary key: Reg\_no
- Foreign keys:None

### 6.Truck:

- Truck [Reg\_no : VARCHAR(50) , Load\_Capacity: DECIMAL(10,2),Trailer\_Type: VARCHAR(50)]
- Primary key: Reg\_no
- Foreign keys:
  - Reg\_no → Vehicle.Reg\_no

### 7.MotorCycle:

- MotorCycle [Reg\_no : VARCHAR(50) , Engine\_Type: VARCHAR(50)]
- Primary key: Reg\_no
- Foreign keys:
  - Reg\_no → Vehicle.Reg\_no

#### 8.Drives:

- Drives [Pid : INT, Reg\_no : VARCHAR(50)]
- Primary key: (Pid,Reg\_no)
- Foreign keys:
  - Pid → Postman.Pid
  - Reg\_no → Vehicle.Reg\_no

#### 9.Telephones:

- Telephones [Cid : INT, Telephone : VARCHAR(20)]
- Primary key: (Cid,Telephone)
- Foreign keys:
  - Cid → Customer.Cid

\*Based on Part 1 feedback, foreign keys have been explicitly included for all tables in the BCNF-normalized schema.

#### Query Demonstration:

##### 1.JOIN Query:

**Explanation:** Join query combines rows from the Parcel and Customer tables based on the Customer ID to retrieve the customer's name for each parcel within the Parcel delivery system. This function allows administration staff to identify which customer ordered a specific parcel.

##### Code:

```
SELECT P.Id,P.Cid,C.F_Name,C.L_Name,P.Status,P.Type
FROM Parcel P, Customer C
WHERE P.Cid=C.Cid;
```

##### Query Result:

| Id | Cid | F_Name | L_Name  | Status     | Type     |
|----|-----|--------|---------|------------|----------|
| 1  | 1   | Jake   | Spencer | In Transit | Standard |
| 2  | 1   | Jake   | Spencer | Delivered  | Express  |
| 3  | 2   | Ruby   | Danson  | Picked Up  | Standard |
| 4  | 2   | Ruby   | Danson  | In Transit | Fragile  |
| 5  | 1   | Jake   | Spencer | Delivered  | Express  |

**Before:** The status of all tables affected by this query before the execution.

i.Customer table:

| <div><div><div>←</div><div>T</div><div>→</div></div></div> |                                                                                   |      |                                                                                        | Cid                                                                                      | F_Name | L_Name | Address |                                            |
|------------------------------------------------------------|-----------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|---------|--------------------------------------------|
| <input type="checkbox"/>                                   |  | Edit |  Copy |  Delete | 1      | Jake   | Spencer | 24/2, Green Tree road, Kallangur, Brisbane |
| <input type="checkbox"/>                                   |  | Edit |  Copy |  Delete | 2      | Ruby   | Danson  | 35/2, Penbrook street, Sydney              |
| <input type="checkbox"/>                                   |  | Edit |  Copy |  Delete | 3      | Elsie  | Kim     | 71/3, Glenview, Perth                      |
| <input type="checkbox"/>                                   |  | Edit |  Copy |  Delete | 4      | Jamie  | Kang    | 42/2, Byron Bay, Sydney                    |
| <input type="checkbox"/>                                   |  | Edit |  Copy |  Delete | 5      | Emma   | Stone   | 65/4, Birch street, Adelaide               |

## 2. Aggregation Query:

**Explanation:** An aggregation query uses functions like MIN, MAX, AVG, or COUNT to summarise data about parcels in the Parcel delivery system. This function allows staff to calculate the total weight, average weight, and total number of parcels, enabling them to assess overall load and volume of parcels for planning operations and allocation of resources.

### Code:

```
SELECT SUM(Weight) AS Total_Weight, AVG(Weight) AS Avg_Weight, COUNT(*) AS Parcel_Count  
FROM Parcel;
```

### Query Result:

| Total_Weight | Avg_Weight | Parcel_Count |
|--------------|------------|--------------|
| 11.50        | 2.300000   | 5            |

**Before:** The status of all tables affected by this query before the execution.

i.Parcel table:

| <div><div><div></div><div></div><div></div></div></div> |                                   |                 |                                   |                 | <div>▼</div>                      | <div>Id</div>     | <div>Cid</div> | <div>Status</div> | <div>Type</div>       | <div>Weight</div>   | <div>Money</div> | <div>Wid</div>   | <div>Pid</div> |              |
|---------------------------------------------------------|-----------------------------------|-----------------|-----------------------------------|-----------------|-----------------------------------|-------------------|----------------|-------------------|-----------------------|---------------------|------------------|------------------|----------------|--------------|
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> | <div>Edit</div> | <div><div><div></div></div></div> | <div>Copy</div> | <div><div><div></div></div></div> | <div>Delete</div> | <div>1</div>   | <div>1</div>      | <div>In Transit</div> | <div>Standard</div> | <div>2.50</div>  | <div>25.00</div> | <div>1</div>   | <div>1</div> |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> | <div>Edit</div> | <div><div><div></div></div></div> | <div>Copy</div> | <div><div><div></div></div></div> | <div>Delete</div> | <div>2</div>   | <div>1</div>      | <div>Delivered</div>  | <div>Express</div>  | <div>1.20</div>  | <div>40.00</div> | <div>2</div>   | <div>2</div> |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> | <div>Edit</div> | <div><div><div></div></div></div> | <div>Copy</div> | <div><div><div></div></div></div> | <div>Delete</div> | <div>3</div>   | <div>2</div>      | <div>Picked Up</div>  | <div>Standard</div> | <div>3.00</div>  | <div>30.00</div> | <div>1</div>   | <div>3</div> |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> | <div>Edit</div> | <div><div><div></div></div></div> | <div>Copy</div> | <div><div><div></div></div></div> | <div>Delete</div> | <div>4</div>   | <div>2</div>      | <div>In Transit</div> | <div>Fragile</div>  | <div>0.80</div>  | <div>15.00</div> | <div>2</div>   | <div>4</div> |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> | <div>Edit</div> | <div><div><div></div></div></div> | <div>Copy</div> | <div><div><div></div></div></div> | <div>Delete</div> | <div>5</div>   | <div>1</div>      | <div>Delivered</div>  | <div>Express</div>  | <div>4.00</div>  | <div>50.00</div> | <div>1</div>   | <div>5</div> |

**After:** The status of all tables affected by this query after the execution.

i.Parcel table:

| <div><div>↩</div><div>T</div><div>→</div></div> |             |                 |             |                 | <div>▼</div> | <div>Id</div>     | <div>Cid</div> | <div>Status</div> | <div>Type</div>       | <div>Weight</div>   | <div>Money</div> | <div>Wid</div>   | <div>Pid</div> |              |
|-------------------------------------------------|-------------|-----------------|-------------|-----------------|--------------|-------------------|----------------|-------------------|-----------------------|---------------------|------------------|------------------|----------------|--------------|
| <div><input type="checkbox"/></div>             | <div></div> | <div>Edit</div> | <div></div> | <div>Copy</div> | <div></div>  | <div>Delete</div> | <div>1</div>   | <div>1</div>      | <div>In Transit</div> | <div>Standard</div> | <div>2.50</div>  | <div>25.00</div> | <div>1</div>   | <div>1</div> |
| <div><input type="checkbox"/></div>             | <div></div> | <div>Edit</div> | <div></div> | <div>Copy</div> | <div></div>  | <div>Delete</div> | <div>2</div>   | <div>1</div>      | <div>Delivered</div>  | <div>Express</div>  | <div>1.20</div>  | <div>40.00</div> | <div>2</div>   | <div>2</div> |
| <div><input type="checkbox"/></div>             | <div></div> | <div>Edit</div> | <div></div> | <div>Copy</div> | <div></div>  | <div>Delete</div> | <div>3</div>   | <div>2</div>      | <div>Picked Up</div>  | <div>Standard</div> | <div>3.00</div>  | <div>30.00</div> | <div>1</div>   | <div>3</div> |
| <div><input type="checkbox"/></div>             | <div></div> | <div>Edit</div> | <div></div> | <div>Copy</div> | <div></div>  | <div>Delete</div> | <div>4</div>   | <div>2</div>      | <div>In Transit</div> | <div>Fragile</div>  | <div>0.80</div>  | <div>15.00</div> | <div>2</div>   | <div>4</div> |
| <div><input type="checkbox"/></div>             | <div></div> | <div>Edit</div> | <div></div> | <div>Copy</div> | <div></div>  | <div>Delete</div> | <div>5</div>   | <div>1</div>      | <div>Delivered</div>  | <div>Express</div>  | <div>4.00</div>  | <div>50.00</div> | <div>1</div>   | <div>5</div> |

The table remains unchanged.

### 3.Aggregation with GROUP BY Query:

**Explanation:** This query groups data by warehouse and applies the COUNT function to determine the number of parcels stored in each warehouse within the Parcel delivery system. This function allows staff to monitor warehouse inventory levels, ensuring efficient storage and distribution management.

#### Code:

```
SELECT W.Wid, W.Name, COUNT(P.Id) AS Parcel_Count
FROM WareHouse W
LEFT JOIN Parcel P ON W.Wid = P.Wid
GROUP BY W.Wid, W.Name;
```

#### Query Result:

| Wid | Name          | Parcel_Count |
|-----|---------------|--------------|
| 1   | Brisbane Hub  | 3            |
| 2   | Adelaide Hub  | 2            |
| 3   | Sydney Hub    | 0            |
| 4   | Perth Hub     | 0            |
| 5   | Melbourne Hub | 0            |

**Before:** The status of all tables affected by this query before the execution.

i.Parcel table:

|  |  |  |  | 1 | 1 | In Transit | Standard | 2.50 | 25.00 | 1 | 1 |
|--|--|--|--|---|---|------------|----------|------|-------|---|---|
|  |  |  |  | 2 | 1 | Delivered  | Express  | 1.20 | 40.00 | 2 | 2 |
|  |  |  |  | 3 | 2 | Picked Up  | Standard | 3.00 | 30.00 | 1 | 3 |
|  |  |  |  | 4 | 2 | In Transit | Fragile  | 0.80 | 15.00 | 2 | 4 |
|  |  |  |  | 5 | 1 | Delivered  | Express  | 4.00 | 50.00 | 1 | 5 |

ii.WareHouse table:

| <div><div><div><div></div><div></div><div></div></div></div><div></div></div> |                                        |                                        |                                          | Wid | Name          | Address                               |
|-------------------------------------------------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|-----|---------------|---------------------------------------|
| <div><div><div></div></div></div>                                             | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 1   | Brisbane Hub  | 300,Peach street,Bowen Hills,Brisbane |
| <div><div><div></div></div></div>                                             | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 2   | Adelaide Hub  | 352,Depot lane,Adelaide               |
| <div><div><div></div></div></div>                                             | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 3   | Sydney Hub    | 200,Industrial Road,Sydney            |
| <div><div><div></div></div></div>                                             | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 4   | Perth Hub     | 402,Freight Road,Perth                |
| <div><div><div></div></div></div>                                             | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 5   | Melbourne Hub | 502,Logistics avenue,Melbourne        |

**After:** The status of all tables affected by this query after the execution.

i.Parcel table:

| <div><div></div><div></div><div></div></div> |                                                                                                   |                                                                                                   |                                                                                                     | <div>▼</div> | <div></div>  | <div></div>           | <div></div>         | <div></div>     | <div></div>      | <div></div>  | <div></div>  | <div></div> |
|----------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------|--------------|-----------------------|---------------------|-----------------|------------------|--------------|--------------|-------------|
| <div><input type="checkbox"/></div>          | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> | <div></div>  | <div></div>  | <div></div>           | <div></div>         | <div></div>     | <div></div>      | <div></div>  | <div></div>  | <div></div> |
| <div><input type="checkbox"/></div>          | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> | <div>1</div> | <div>1</div> | <div>In Transit</div> | <div>Standard</div> | <div>2.50</div> | <div>25.00</div> | <div>1</div> | <div>1</div> |             |
| <div><input type="checkbox"/></div>          | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> | <div>2</div> | <div>1</div> | <div>Delivered</div>  | <div>Express</div>  | <div>1.20</div> | <div>40.00</div> | <div>2</div> | <div>2</div> |             |
| <div><input type="checkbox"/></div>          | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> | <div>3</div> | <div>2</div> | <div>Picked Up</div>  | <div>Standard</div> | <div>3.00</div> | <div>30.00</div> | <div>1</div> | <div>3</div> |             |
| <div><input type="checkbox"/></div>          | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> | <div>4</div> | <div>2</div> | <div>In Transit</div> | <div>Fragile</div>  | <div>0.80</div> | <div>15.00</div> | <div>2</div> | <div>4</div> |             |
| <div><input type="checkbox"/></div>          | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> | <div>5</div> | <div>1</div> | <div>Delivered</div>  | <div>Express</div>  | <div>4.00</div> | <div>50.00</div> | <div>1</div> | <div>5</div> |             |

ii.WareHouse table:

|  |                                                                                   |                                                                                   |                                                                                   |     |               |                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----|---------------|---------------------------------------|
| <input type="checkbox"/>                                                          |  |  |  | Wid | Name          | Address                               |
| <input type="checkbox"/>                                                          |  |  |  | 1   | Brisbane Hub  | 300,Peach street,Bowen Hills,Brisbane |
| <input type="checkbox"/>                                                          |  |  |  | 2   | Adelaide Hub  | 352,Depot lane,Adelaide               |
| <input type="checkbox"/>                                                          |  |  |  | 3   | Sydney Hub    | 200,Industrial Road,Sydney            |
| <input type="checkbox"/>                                                          |  |  |  | 4   | Perth Hub     | 402,Freight Road,Perth                |
| <input type="checkbox"/>                                                          |  |  |  | 5   | Melbourne Hub | 502,Logistics avenue,Melbourne        |

The tables remain unchanged.

#### 4.Division Query:

**Explanation:** The division query identifies postmen who have delivered parcels of every type (e.g., 'Standard','Express','Fragile') within the Parcel delivery system. This function allows administration staff in recognising high-performing postmen and nominate them for a bonus.

#### Code:

```
SELECT P.Pid, P.Name
FROM Postman P
WHERE NOT EXISTS (
    SELECT DISTINCT Type
    FROM Parcel P1
    WHERE NOT EXISTS (
        SELECT 1
        FROM Parcel P2
        WHERE P2.Pid = P.Pid
        AND P2.Type = P1.Type
    )
);
```

### Query Result:

| Pid | Name |
|-----|------|
|-----|------|

The query returned no rows, indicating that no postman has delivered all types of parcels.

**Before:** The status of all tables affected by this query before the execution.

i.Parcel table:

| <div>← T →</div>                    |                                                                                                   |                                                                                                   |                                                                                                     |  | <div>▼</div> | <div>Id</div> | <div>Cid</div>        | <div>Status</div>   | <div>Type</div> | <div>Weight</div> | <div>Money</div> | <div>Wid</div> | <div>Pid</div> |
|-------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--|--------------|---------------|-----------------------|---------------------|-----------------|-------------------|------------------|----------------|----------------|
| <div><input type="checkbox"/></div> | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> |  | <div>1</div> | <div>1</div>  | <div>In Transit</div> | <div>Standard</div> | <div>2.50</div> | <div>25.00</div>  | <div>1</div>     | <div>1</div>   |                |
| <div><input type="checkbox"/></div> | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> |  | <div>2</div> | <div>1</div>  | <div>Delivered</div>  | <div>Express</div>  | <div>1.20</div> | <div>40.00</div>  | <div>2</div>     | <div>2</div>   |                |
| <div><input type="checkbox"/></div> | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> |  | <div>3</div> | <div>2</div>  | <div>Picked Up</div>  | <div>Standard</div> | <div>3.00</div> | <div>30.00</div>  | <div>1</div>     | <div>3</div>   |                |
| <div><input type="checkbox"/></div> | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> |  | <div>4</div> | <div>2</div>  | <div>In Transit</div> | <div>Fragile</div>  | <div>0.80</div> | <div>15.00</div>  | <div>2</div>     | <div>4</div>   |                |
| <div><input type="checkbox"/></div> | <div> Edit</div> | <div> Copy</div> | <div> Delete</div> |  | <div>5</div> | <div>1</div>  | <div>Delivered</div>  | <div>Express</div>  | <div>4.00</div> | <div>50.00</div>  | <div>1</div>     | <div>5</div>   |                |

ii.Postman table:

|                          |  |      |  |      | Pid | Name   | Contact                    |
|--------------------------|--|------|--|------|-----|--------|----------------------------|
| <input type="checkbox"/> |  | Edit |  | Copy |     | Delete | 1 Tommy Lee 0411-111-111   |
| <input type="checkbox"/> |  | Edit |  | Copy |     | Delete | 2 Jackie Chan 0422-222-222 |
| <input type="checkbox"/> |  | Edit |  | Copy |     | Delete | 3 Brad Pitt 0433-333-333   |
| <input type="checkbox"/> |  | Edit |  | Copy |     | Delete | 4 David Wiese 0444-444-444 |
| <input type="checkbox"/> |  | Edit |  | Copy |     | Delete | 5 Steve Smith 0455-555-555 |

**After:** The status of all tables affected by this query after the execution.

i.Parcel table:

| <div><div><div></div><div></div><div></div></div></div> |                                        |                                        |                                          |   | <div>Id</div> | <div>Cid</div> | <div>Status</div> | <div>Type</div> | <div>Weight</div> | <div>Money</div> | <div>Wid</div> | <div>Pid</div> |
|---------------------------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|---|---------------|----------------|-------------------|-----------------|-------------------|------------------|----------------|----------------|
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 1 | 1             | In Transit     | Standard          | 2.50            | 25.00             | 1                | 1              |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 2 | 1             | Delivered      | Express           | 1.20            | 40.00             | 2                | 2              |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 3 | 2             | Picked Up      | Standard          | 3.00            | 30.00             | 1                | 3              |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 4 | 2             | In Transit     | Fragile           | 0.80            | 15.00             | 2                | 4              |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 5 | 1             | Delivered      | Express           | 4.00            | 50.00             | 1                | 5              |                |

ii.Postman table:

| <div><div><div></div><div></div><div></div></div></div> |                                                                                   |      |                                                                                   |      | Pid                                                                               | Name   | Contact |             |              |
|---------------------------------------------------------|-----------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------|--------|---------|-------------|--------------|
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 1       | Tommy Lee   | 0411-111-111 |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 2       | Jackie Chan | 0422-222-222 |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 3       | Brad Pitt   | 0433-333-333 |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 4       | David Wiese | 0444-444-444 |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 5       | Steve Smith | 0455-555-555 |

The tables remain unchanged.

### 5.Update Query:

**Explanation:** An update operation modifies existing rows in the Parcel table to change the status of parcels stored in WareHouse 1 to 'Picked Up' within the Parcel delivery system. This function allows staff to upgrade parcel statuses as they are collected from shippers, maintaining accurate tracking information.

#### Code:

```
UPDATE Parcel
```

```
SET Status="Picked Up"
```

```
WHERE Wid=1;
```

**Before:** The status of all tables affected by this query before the execution.

i.Parcel table:

| <div><div><div></div><div></div><div></div></div></div> |                                        |                                        |                                          |   |   | <div>Id</div> | <div>Cid</div> | <div>Status</div> | <div>Type</div> | <div>Weight</div> | <div>Money</div> | <div>Wid</div> | <div>Pid</div> |
|---------------------------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|---|---|---------------|----------------|-------------------|-----------------|-------------------|------------------|----------------|----------------|
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 1 | 1 | In Transit    | Standard       | 2.50              | 25.00           | 1                 | 1                |                |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 2 | 1 | Delivered     | Express        | 1.20              | 40.00           | 2                 | 2                |                |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 3 | 2 | Picked Up     | Standard       | 3.00              | 30.00           | 1                 | 3                |                |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 4 | 2 | In Transit    | Fragile        | 0.80              | 15.00           | 2                 | 4                |                |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 5 | 1 | Delivered     | Express        | 4.00              | 50.00           | 1                 | 5                |                |                |

**After:** The status of all tables affected by this query after the execution.

i. Parcel table:

| <div><div><div></div><div></div><div></div></div></div> |                                        |                                        |                                          |   | <div>Id</div> | <div>Cid</div> | <div>Status</div> | <div>Type</div> | <div>Weight</div> | <div>Money</div> | <div>Wid</div> | <div>Pid</div> |
|---------------------------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|---|---------------|----------------|-------------------|-----------------|-------------------|------------------|----------------|----------------|
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 1 | 1             | Picked Up      | Standard          | 2.50            | 25.00             | 1                | 1              |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 2 | 1             | Delivered      | Express           | 1.20            | 40.00             | 2                | 2              |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 3 | 2             | Picked Up      | Standard          | 3.00            | 30.00             | 1                | 3              |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 4 | 2             | In Transit     | Fragile           | 0.80            | 15.00             | 2                | 4              |                |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 5 | 1             | Picked Up      | Express           | 4.00            | 50.00             | 1                | 5              |                |

## 6. Delete operation with Cascade:

**Explanation:** A Delete with Cascade Operation removes a customer tuple from the Customer table and automatically deletes all related parcel and telephone number records in the Parcel and Telephones table that reference the deleted customer via foreign keys within the Parcel delivery system. This function allows staff to remove a customer and their associated delivery records ( e.g., when a customer is no longer active), ensuring the database remains up to date.

**Code:**

```
DELETE FROM Customer
```

```
WHERE Cid=1;
```

**Before:** The status of all tables affected by this query before the execution.

i.Customer table:

| <div><div><div></div><div></div><div></div></div></div> |                                                                                     |      |                                                                                     |      | Cid                                                                                 | F_Name | L_Name | Address |         |                                         |
|---------------------------------------------------------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|--------|--------|---------|---------|-----------------------------------------|
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 1      | Jake    | Spencer | 24/2,Green Tree road,Kallangur,Brisbane |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 2      | Ruby    | Danson  | 35/2,Penbrook street,Sydney             |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 3      | Elsie   | Kim     | 71/3,Glenview,Perth                     |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 4      | Jamie   | Kang    | 42/2,Byron Bay,Sydney                   |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 5      | Emma    | Stone   | 65/4,Birch street,Adelaide              |

ii.Parcel table:

| <div><div><div></div><div></div><div></div></div></div> |                                        |                                        |                                          |  | Id | Cid | Status     | Type     | Weight | Money | Wid | Pid |
|---------------------------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|--|----|-----|------------|----------|--------|-------|-----|-----|
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete |  | 1  | 1   | Picked Up  | Standard | 2.50   | 25.00 | 1   | 1   |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete |  | 2  | 1   | Delivered  | Express  | 1.20   | 40.00 | 2   | 2   |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete |  | 3  | 2   | Picked Up  | Standard | 3.00   | 30.00 | 1   | 3   |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete |  | 4  | 2   | In Transit | Fragile  | 0.80   | 15.00 | 2   | 4   |
| <div><div><div></div></div></div>                       | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete |  | 5  | 1   | Picked Up  | Express  | 4.00   | 50.00 | 1   | 5   |

iii.Telephones table:

| <div><div><div></div><div></div><div></div></div></div> |                                                                                   |      |                                                                                   |      |                                                                                   | Cid    | Telephone      |
|---------------------------------------------------------|-----------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------|--------|----------------|
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 1 0411-123-456 |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 2 0411-654-321 |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 2 0422-987-654 |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 3 0433-456-789 |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 4 0444-321-987 |

**After:** The status of all tables affected by this query after the execution.

i.Customer table:

| <div><div><div></div><div></div><div></div></div></div> |                                                                                    |      |                                                                                    |      | Cid                                                                                | F_Name | L_Name  | Address |                               |
|---------------------------------------------------------|------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------|--------|---------|---------|-------------------------------|
| <input type="checkbox"/>                                |   | Edit |   | Copy |   | Delete | 2 Ruby  | Danson  | 35/2, Penbrook street, Sydney |
| <input type="checkbox"/>                                |   | Edit |   | Copy |   | Delete | 3 Elsie | Kim     | 71/3, Glenview, Perth         |
| <input type="checkbox"/>                                |   | Edit |   | Copy |   | Delete | 4 Jamie | Kang    | 42/2, Byron Bay, Sydney       |
| <input type="checkbox"/>                                |  | Edit |  | Copy |  | Delete | 5 Emma  | Stone   | 65/4, Birch street, Adelaide  |

ii.Parcel table:

| <div><div><div>←</div><div>T</div><div>→</div></div><div></div></div> |                                        |                                        |                                          |   |   | <div>Id</div> | <div>Cid</div> | <div>Status</div> | <div>Type</div> | <div>Weight</div> | <div>Money</div> | <div>Wid</div> | <div>Pid</div> |
|-----------------------------------------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|---|---|---------------|----------------|-------------------|-----------------|-------------------|------------------|----------------|----------------|
| <div><div><div></div></div></div>                                     | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 3 | 2 | Picked Up     | Standard       | 3.00              | 30.00           | 1                 | 3                |                |                |
| <div><div><div></div></div></div>                                     | <div><div><div></div></div></div> Edit | <div><div><div></div></div></div> Copy | <div><div><div></div></div></div> Delete | 4 | 2 | In Transit    | Fragile        | 0.80              | 15.00           | 2                 | 4                |                |                |

iii.Telephones table:

|  |                                                                                     |      |                                                                                     |      |                                                                                     | Cid    | Telephone      |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|--------|----------------|
| <input type="checkbox"/>                                                            |  | Edit |  | Copy |  | Delete | 2 0411-654-321 |
| <input type="checkbox"/>                                                            |  | Edit |  | Copy |  | Delete | 2 0422-987-654 |
| <input type="checkbox"/>                                                            |  | Edit |  | Copy |  | Delete | 3 0433-456-789 |
| <input type="checkbox"/>                                                            |  | Edit |  | Copy |  | Delete | 4 0444-321-987 |

\*Grammarly was used to proofread the Query Explanations for grammar and clarity.